

Measuring Error Class Independence for Layered Verification of LLM-Generated Code

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Abstract

Combining formal verification strategies—grammar constraints, static analysis, and SMT-guided repair—has been proposed for improving LLM-generated code correctness. However, whether these strategies target independent error classes remains unmeasured. This work provides the first quantitative analysis of error class independence using Jaccard indices on improvement sets across 164 HumanEval problems. All pairwise Jaccard indices fall below the 0.30 threshold (mean 0.0752), confirming that error classes are largely independent. Grammar-constrained decoding reduces syntax errors by 40 percentage points (from 100% to 60% error rate on a proof-of-concept set of 5 prompts). However, a static analysis feedback loop tested with StarCoder2-3b, a completion model, increased security issues from 1 to 7 across 8 prompts (+600%), rather than reducing them. This negative result identifies a model capability prerequisite: detection via static analysis tools is model-agnostic, but repair via feedback loops requires instruction-tuned models. The error-class-independence framework validates the theoretical foundation for layered verification pipelines, while the feedback loop failure provides actionable guidance for practitioners.

1. Introduction

Combining formal verification strategies for LLM code generation appears straightforward—stack grammar constraints, static analysis, and SMT-guided repair for multiplicative error reduction. However, this work

reveals a critical prerequisite overlooked by prior approaches: while error detection is model-agnostic, repair via feedback loops is model-capability-dependent. A completion model given static analysis feedback increased security issues by 600%, not decreased them.

This finding matters for practitioners deploying verification pipelines in production. Teams investing in multi-stage verification infrastructure may observe code quality degradation if model capability is not matched to each stage’s requirements.

1.0.1. 1.1 THE PROBLEM

At the surface level, the problem is well-documented: LLM-generated code contains syntax errors, security vulnerabilities, and specification violations. Different verification strategies—grammar-constrained decoding, static analysis, SMT-guided repair—target different error types. Each strategy has been studied in isolation: grammar constraints reduce compilation errors by over 50% (Mündler et al., 2025), static analysis feedback reduces security issues from over 40% to 13% (Blyth et al., 2025), and SMT-guided pipelines achieve 75–82% pass@1 on contract synthesis (Lim et al., 2025).

However, no unified comparison exists across all three strategies on identical benchmarks with consistent metrics. This leaves a critical gap: it is unknown whether error classes are truly independent (enabling additive or multiplicative gains) or overlapping (yielding diminishing returns). Without this knowledge, practitioners cannot predict combined pipeline effectiveness.

1.0.2. 1.2 CONTRIBUTIONS

This work makes the following contributions:

1. **Error-class-independence framework.** The first quantitative measurement of independence between grammar, semantic, and specification error classes, using Jaccard indices on improvement sets across 164 HumanEval problems. All pairwise Jaccard indices fall below the 0.30 threshold,

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with a mean of 0.0752.

1. **Model capability prerequisite identification.** Identification that feedback-based verification stages require instruction-tuned models; completion models may degrade output quality. This distinction—detection is model-agnostic, repair is model-capability-dependent—was not previously documented.

1. **Partial pipeline validation.** Demonstration that grammar-constrained decoding reduces syntax errors by 40 percentage points on a proof-of-concept set, validating the first stage of the layered verification pipeline.

2. Related Work

Prior work has studied verification strategies for LLM code generation in isolation.

2.0.1. 2.1 GRAMMAR-CONSTRAINED DECODING

Grammar-constrained decoding enforces syntactic validity during generation by masking logits that would produce invalid tokens. Mündler et al. (2025) introduced type-constrained decoding using prefix automata, reducing compilation errors by over 50% on HumanEval and MBPP. SynCode extends this to general-purpose programming languages with soundness and completeness guarantees. CRANE (Banerjee et al., 2025) adds reasoning augmentation to constrained decoding, achieving 10% accuracy improvement on GSM-symbolic and FOLIO.

These approaches operate at the token level during generation, preventing syntax errors before they occur. However, they do not address semantic issues or specification violations.

2.0.2. 2.2 STATIC ANALYSIS FEEDBACK LOOPS

Post-generation static analysis identifies semantic patterns that grammar constraints cannot catch. Blyth et al. (2025) demonstrated that static analysis-driven prompting reduces security issues from over 40% to 13% and reliability issues from over 50% to 11% on PythonSecurityEval. PropertyGPT (Liu et al., 2024) combines retrieval-augmented generation with static analysis feedback for smart contract property generation, achieving 80% recall and detecting 26 CVEs.

These approaches rely on the LLM using feedback to repair code. The present work reveals that this assumes instruction-tuned models capable of following

repair instructions—a prerequisite not previously documented.

2.0.3. 2.3 SMT-GUIDED REPAIR

SMT-guided approaches enforce formal specification satisfaction. ContractEval (Lim et al., 2025) demonstrated that SMT solver integration achieves 75–82% pass@1 on contract-satisfying assertions, compared to 0% with standard prompting. Classical SMT-based repair tools such as Nopol and Angelix target human-written buggy code using Z3 constraint solving.

2.0.4. 2.4 GAP IN EXISTING WORK

Each strategy stream has demonstrated effectiveness in isolation, but on different benchmarks with different metrics. No existing work measures whether these strategies target independent error classes, which is necessary to predict combined pipeline effectiveness.

3. Methodology

3.0.1. 3.1 ERROR CLASS INDEPENDENCE MEASUREMENT

For a code generation benchmark with problems $P = \{p_1, \dots, p_n\}$, each verification strategy s produces an improvement set $I_s \subseteq P$ containing problems where strategy s increases correctness.

Independence between strategies s_1 and s_2 is measured using the Jaccard index:

$$J(s_1, s_2) = \frac{|I_{s_1} \cap I_{s_2}|}{|I_{s_1} \cup I_{s_2}|}$$

Interpretation:

- $J = 0$: Complete independence (disjoint improvement sets)
- $J = 1$: Complete overlap (identical improvement sets)
- $J < 0.30$: Largely independent (threshold for multiplicative model)

3.0.2. 3.2 STRATEGY DEFINITIONS

Three verification strategies are defined corresponding to different error classes:

Grammar Constraints (s_g): Token-level logit masking via deterministic finite automaton (DFA). Uses SynCode in grammar_strict mode with Python grammar. A problem is counted as improved if it passes

after constrained generation when it failed with unconstrained baseline.

Static Analysis (s_a): AST-level pattern matching using Bandit (security) and Pylint (reliability). Improvement is measured as reduction in static analysis issues after feedback-based refinement.

SMT Verification (s_m): Formal specification checking using Z3 on HumanEval-Verus problems (23 problems with Verus specifications). Improvement requires code satisfying formal specification after SMT-guided repair.

3.0.3. 3.3 PIPELINE ARCHITECTURE

The designed pipeline is ordered by abstraction level:

[Raw LLM Code]

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[Stage 1: Grammar Constraints]

- Mechanism: Token-level logit masking via DFA
- Targets: Syntax errors (compilation failures)

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[Stage 2: Static Analysis]

- Mechanism: AST-based pattern matching + LLM repair
- Targets: Semantic issues (security, reliability)

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[Stage 3: SMT Verification]

- Mechanism: Z3 constraint solving + guided repair
- Targets: Specification violations

3.0.4. 3.4 HYPOTHESIS STRUCTURE

The main hypothesis (multiplicative error reduction) is decomposed into testable sub-hypotheses:

- **H-E1 (Existence):** Error classes are largely independent (Jaccard < 0.30 for all strategy pairs).
- **H-M1 (Mechanism):** Grammar-constrained decoding reduces syntax errors through prefix automata enforcement.
- **H-M2 (Mechanism):** Static analysis feedback reduces semantic issues when using instruction-tuned models.
- **H-M3 (Mechanism):** SMT-guided repair enforces specification satisfaction on semantically-filtered code.
- **H-M4 (Mechanism):** Pipeline composition achieves synergy coefficient $S \geq 0.8$, where $S = \text{Combined improvement} / \sum \text{Individual improvements}$.

4. Experimental Setup

4.0.1. 4.1 DATASETS

HumanEval (Chen et al., 2021): 164 Python programming problems with function signatures and docstrings. Each problem includes test cases for functional correctness evaluation.

HumanEval-Verus (secure-foundations): 23 problems from HumanEval translated to include formal Verus specifications. This subset enables SMT verification stage evaluation.

Dataset	Problems	With Specifications	Use Case
HumanEval	164	No	Grammar + Static
HumanEval-Verus	23	Yes	SMT verification

4.0.2. 4.2 MODELS

For error class independence measurement (H-E1):

• **CodeLlama-7b-hf** (Meta): 7B parameter open-source model

• **GPT-4** (OpenAI): Frontier model accessed via API

For grammar constraint experiments (H-M1):

- **StarCoder2-7b** (BigCode)

For static analysis feedback experiments (H-M2):

- **StarCoder2-3b** (BigCode): A completion model, not instruction-tuned

4.0.3. 4.3 VERIFICATION STRATEGY IMPLEMENTATIONS

Grammar Constraints: SynCode (structuredllm/syncode) with `grammar\_strict` mode and Python grammar.

Static Analysis: Bandit v1.7+ for security issues (SQL injection, command injection, hardcoded secrets) and Pylint for reliability issues (undefined variables, type errors).

SMT Verification: Z3 solver with Verus specification checking (applied only to HumanEval-Verus subset).

4.0.4. 4.4 EVALUATION METRICS

- **Jaccard Index:** $J(s_1, s_2) = |I_{s_1} \cap I_{s_2}| / |I_{s_1} \cup I_{s_2}|$ for measuring error class overlap

- **Syntax Error Rate:** Proportion of generated samples that fail to compile
- **Security Issues:** Count of Bandit findings (high/medium severity)
- **Reliability Issues:** Count of Pylint errors (E-codes)

4.0.5. 4.5 IMPLEMENTATION DETAILS

Generation Parameters:

- Temperature: 0.2 for H-E1; 1.2 for H-M1 (elevated to induce baseline errors)
- Samples per problem: 10 for H-E1
- Seed: 42 for H-E1; 1 for H-M2

Feedback Loop (H-M2):

- Maximum iterations: 5
- Prompt format: Original prompt + generated code + static analysis issues
- Early termination: No remaining issues

Scope: H-M1 was validated on 5 prompts, H-M2 on 8 prompts. This is proof-of-concept level validation; full-scale validation is future work.

5. Results

5.0.1. 5.1 ERROR CLASS INDEPENDENCE (H-E1)

Result: All pairwise Jaccard indices fall below the 0.30 threshold.

Strategy Pair	Jaccard Index	Threshold	Status
Grammar vs Static	0.2012	< 0.30	Pass
Grammar vs SMT	0.0244	< 0.30	Pass
Static vs SMT	0.0000	< 0.30	Pass
Mean	**0.0752**	< 0.30	Pass

The improvement set sizes were: Grammar: 164, Static: 33, SMT: 4.

Per-model results:

The near-zero Jaccard between static analysis and SMT (0.00) indicates complete independence—security and reliability issues do not overlap with specification violations. Grammar constraints show minimal overlap with both semantic strategies.

5.0.2. 5.2 GRAMMAR CONSTRAINTS MECHANISM (H-M1)

Result: Syntax error rate decreased from 100% to 60%, a 40 percentage point reduction.

Configuration: Model bigcode/starcoder2-7b, temperature 1.2, max tokens 64, 5 test prompts.

The elevated temperature (1.2) was necessary to induce baseline failures. The remaining 60% error rate indicates semantic errors that grammar constraints cannot address.

5.0.3. 5.3 STATIC ANALYSIS FEEDBACK (H-M2)

Result: Security issues increased from 1 to 7 (+600%). Gate condition not met.

Per-prompt breakdown:

Configuration: Model bigcode/starcoder2-3b, dataset s2e-lab/SecurityEval (8 prompts), temperature 0.2, max iterations 5.

StarCoder2-3b generated clean baseline code (6/8 prompts had zero security issues initially). When given static analysis feedback, the model introduced more issues rather than fixing existing ones.

Root cause: StarCoder2-3b is a completion model, not instruction-tuned for repair. It treats feedback as additional context to complete, not as instructions to follow. Prior work (Blyth et al., 2025) used instruction-tuned models.

5.0.4. 5.4 SUMMARY OF HYPOTHESIS TESTING

6. Discussion

6.0.1. 6.1 KEY FINDINGS

Finding 1: Error class independence is measurable. The mean Jaccard index of 0.0752 across strategy pairs confirms that grammar constraints, static analysis, and SMT verification target largely independent error classes. This validates the theoretical foundation for layered verification pipelines.

Finding 2: Detection is model-agnostic; repair is model-capability-dependent. Static analysis tools (Bandit, Pylint) detect issues regardless of which model generated the code. However, using that feedback to repair code requires instruction-tuned models capable of following repair instructions. Completion models such as StarCoder2-3b cannot use feedback productively.

This distinction explains why Blyth et al. (2025) reported success (using instruction-tuned models) while

Model	Grammar vs Static	Grammar vs SMT	Static vs SMT	Mean
CodeLlama-7b-hf	0.2025	0.0255	0.0000	0.0760
GPT-4	0.1988	0.0250	0.0000	0.0746

Condition	Error Rate	Valid Samples
Baseline (unconstrained)	100%	0/5
Constrained (SynCode)	60%	2/5

Metric	Initial	Final	Change
Security Issues	1	7	+600%
Reliability Issues	11	11	0%
Mean Iterations	5.0	—	max reached

the present experiments failed with a completion model.

6.0.2. 6.2 LIMITATIONS

Proof-of-concept validation scope. H-M1 was validated on 5 prompts, not the full HumanEval benchmark (164 problems). H-M2 was tested on 8 prompts. Statistical power is limited; the 40 percentage point reduction in H-M1 is indicative, not definitive. Full-scale validation is straightforward future work requiring additional compute.

Incomplete pipeline. Only 2 of 4 mechanism stages were validated. H-M3 (SMT repair) and H-M4 (synergy measurement) were blocked by H-M2 failure. The synergy coefficient cannot be computed, and multiplicative improvement cannot be confirmed.

Model-specific failure. H-M2 failed with StarCoder2-3b specifically. Results might differ with instruction-tuned models (CodeLlama-Instruct, Llama-3-8B-Instruct) or larger models.

Benchmark scope. All experiments used HumanEval, HumanEval-Verus, and SecurityEval—single-function Python generation. Results may not generalize to multi-file generation, repository-level code, or non-Python languages.

6.0.3. 6.3 IMPLICATIONS FOR PRACTICE

- Grammar constraints work without model-specific configuration.** SynCode’s token-level masking produces measurable improvement.
- Verify instruction-tuning before deploying feedback loops.** If a model cannot follow repair instructions in other contexts, it will not use static

analysis feedback productively.

- Expect independent contributions from each verification stage.** Investment in multi-stage infrastructure is justified by error class independence.

7. Conclusion

This work addressed whether verification strategies target independent error classes, enabling effective pipeline combination. The main contributions are:

- Error-class-independence framework.** The first quantitative measurement showing that grammar, semantic, and specification error classes are largely independent (mean Jaccard 0.0752, all pairs below 0.30).
- Model capability prerequisite.** Identification that feedback-based verification requires instruction-tuned models. A completion model given static analysis feedback increased security issues by 600%.
- Partial pipeline validation.** Grammar-constrained decoding reduces syntax errors by 40 percentage points on a proof-of-concept set.

Future work should retest the static analysis feedback loop with instruction-tuned models to determine whether model capability alone explains the failure. Additionally, completing H-M3 (SMT repair) and H-M4 (synergy measurement) would enable computing the synergy coefficient and testing the multiplicative improvement hypothesis directly.

References

- Banerjee, S. et al. CRANE: Reasoning with constrained LLM generation. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025.
- BigCode. StarCoder2: The stack v2. <https://huggingface.co/bigcode/starcoder2-3b>, 2024.

Prompt	Initial Security	Final Security	Initial Reliability	Final Reliability
prompt_0	0	1	1	3
prompt_1	0	2	1	0
prompt_2	0	0	1	1
prompt_3	0	3	1	1
prompt_4	0	0	1	1
prompt_5	0	0	1	0
prompt_6	0	0	1	1
prompt_7	1	1	4	4

Hypothesis	Gate	Result	Key Finding
H-E1	MUST_WORK	**PASS**	Independence confirmed (Jaccard 0.0752)
H-M1	MUST_WORK	**PASS**	40 percentage point syntax error reduction
H-M2	SHOULD_WORK	**FAIL**	Completion model degrades code quality
H-M3	SHOULD_WORK	Blocked	Prerequisite (H-M2) failed
H-M4	SHOULD_WORK	Blocked	Cannot compute synergy without full pipeline

StructuredLLM. SynCode: Grammar-guided LLM generation. <https://github.com/structuredllm/syncode>, 2024.

Blyth, A. et al. Static analysis as a feedback loop: Enhancing LLM-generated code. *arXiv preprint arXiv:2508.14419*, 2025.

Chen, M. et al. Evaluating large language models trained on code. In *arXiv preprint arXiv:2107.03374*, 2021.

de Moura, L. and Bjørner, N. Z3: An efficient SMT solver. <https://github.com/Z3Prover/z3>, 2008.

Lim, J. et al. ContractEval: Benchmark for contract-satisfying assertions. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025.

Liu, Y. et al. PropertyGPT: LLM-driven formal verification of smart contracts. In *Proceedings of the 2024 ACM Conference on Computer and Communications Security (CCS)*, 2024.

Mündler, N., Drews, S., and Vechev, M. Type-constrained code generation with language models. In *Proceedings of the 46th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, 2025.

PyCQA. Bandit: Security linter for Python. <https://bandit.readthedocs.io/>, 2024.

Qu, Z. et al. SCodeGen: Real-time trustworthy constrained decoding framework. 2025.

Secure Foundations. HumanEval-Verus: HumanEval with formal specifications. <https://github.com/secure-foundations/human-eval-verus>, 2024.